

1. Fix two positive integers m, n where m and n are relatively prime (meaning $\gcd(m, n) = 1$). Consider the system of congruences

$$\begin{cases} x \equiv a \pmod{m} \\ x \equiv b \pmod{n} \end{cases} \quad (\clubsuit)$$

where a and b are arbitrary integers.

- (a) Prove that if $rm + sn = 1$, then $x = asn + brm$ is a solution to system $\clubsuit$.
- (b) Prove that $\clubsuit$ has a solution for all choices of a and b .
- (c) Fix a solution x_1 to system $\clubsuit$. Show that every element in $[x_1]_{mn}$ is a solution to system $\clubsuit$.
- (d) Fix a solution x_1 to system $\clubsuit$. Show the set of all solutions to $\clubsuit$ is exactly $[x_1]_{mn}$.
Hint: use the fundamental theorem of arithmetic to show that if two relatively prime integers divides some integer, then so does their product.]
- (e) Find **all** integer solutions $x \in \mathbb{Z}$ to the system $\{x \equiv 11 \pmod{72}, x \equiv 30 \pmod{169}\}$.

(a) P.F. Take $x = asn + brm$ into the system

1. into $x \equiv a \pmod{m}$

$$x = asn + brm \equiv a \pmod{m}$$

$$asn \equiv a \pmod{m}$$

Then since $rm + sn = 1$, $a(-rm) \equiv a \pmod{m}$

$$a - arm \equiv a \pmod{m}$$

Since $m \mid arm$

then we have $a \equiv a \pmod{m}$

which is guaranteed by the property of modular congruence.

Similarly, take $x = asn + brm$ into

2. $x \equiv b \pmod{n}$

$$\text{We will have } brm \equiv b \pmod{n}$$

$$b(-sn) \equiv b \pmod{n}$$

$$b - bsn \equiv b \pmod{n}$$

Since $n \mid bsn$

then we have $b \equiv b \pmod{n}$
which is guaranteed by the property
of modular congruence.

Then we have proved the statement by
verifying x is a solution to the system when
 $rm + rs = 1$.

(b). Since $\gcd(m, n) = 1$

By Bezout, there exists $r, s \in \mathbb{Z}$ such that
 $rm + sn = 1$

So by (a), regardless of the choice of a, b ,
 $x = asn + brm$ is a solution to the system.

(c) Fix x_1 as a solution to the system,
Let x be an arbitrary element in $[x_1]_{mn}$

Then $x = x_1 + kmn$ for some $k \in \mathbb{Z}$.

$$\text{then } x = x_1 + kmn \equiv x_1 \equiv a \pmod{m}$$

$$x = x_1 + kmn \equiv x_1 \equiv b \pmod{n}$$

So x is a solution to the system.

Therefore every element in $[x_1]_{mn}$ is a solution
to the system if x_1 is a solution.

(d) Fix x_1 as a solution to the system,

$$\text{So } x_1 = a + gm,$$

$$x_1 = b + fn \text{ for some integers } p, q.$$

Let x be an arbitrary solution to the system.

$$\text{Then } x = a + pm = b + qn$$

$$\text{So } x - x_1 = (p - g)m$$

$$x - x_1 = (q - f)n$$

$$\text{So } m \mid (x - x_1), \quad n \mid (x - x_1)$$

By FTA, $m = p_1 p_2 \dots p_\beta$ for some primes

$$n = q_1 q_2 \dots q_\alpha \text{ for some primes } q_1, \dots, q_\alpha$$

Since $(m, n) = 1$, none of $p_1, \dots, p_\beta$ is the same as any of $q_1, \dots, q_\alpha$

So $(x - x_1) = \gamma p_1 \dots p_\beta q_1 \dots q_\alpha$ for some integer γ , then $x - x_1 = \gamma mn$

Therefore $x = (x_1 + \gamma mn) \in [x_1]_{mn}$

Denote all solutions of the system as S
the set of

Then we just proved $S \subseteq [x_1]_{mn}$.

And By (c) we know $[x_1]_{mn} \subseteq S$

Therefore $S = [x_1]_{mn}$, we have proved the statement.

(e) By Euclidean Algorithm,

$$169 = 2 \times 72 + 25$$

$$72 = 2 \times 25 + 22$$

$$25 = 22 + 3$$

$$22 = 7 \times 3 + 1$$

$$3 = 3 \times 1 + 0$$

$$\text{So } \gcd(169, 72) = 1$$

Then by (d), the set of solution to the system is $[x_1]_{169 \times 72}$, where x_1 is any solution to the system.

$$\begin{aligned} 1 &= 22 - 7 \times (25 - 22) = 8 \times 22 - 7 \times 25 \\ &= 8 \times (72 - 2 \times 25) - 7 \times (169 - 72 \times 2) \\ &= 22 \times 72 - 16 \times 25 - 7 \times 169 \\ &= 22 \times 72 - 16 \times (169 - 2 \times 72) - 7 \times 169 \\ &= 54 \times 72 - 23 \times 169 \end{aligned}$$

$$\text{So let } x_1 = 30 \times 54 \times 72 - 11 \times 23 \times 169$$

$$\begin{aligned} \text{So } x_1 &\equiv -11 \times 23 \times 169 \equiv (-11) \times (54 \times 72 - 1) \\ &\equiv 11 - 54 \times 72 \end{aligned}$$

$$\text{Similarly } x_1 \equiv 30 \pmod{169} \equiv 11 \pmod{72}$$

Therefore x_1 is a solution to the system
 So all the solution to the system consist
 of the set $[x_1]_{mn} = [30 \times 54 \times 72 - 11 \times 23 \times 169]_{72 \times 169}$
 $= [73883]_{12168}$

2. When we define a function on $\mathbb{Z}_n$, we need to check that it is well-defined; many possible "rules" we could think to assign are not well-defined.

(a) Is the assignment

$$\mathbb{Z}_3 \longrightarrow \mathbb{Z}_6$$

$$[a]_3 \longmapsto [a]_6$$

a well-defined function?

(b) Is the assignment

$$\mathbb{Z}_6 \longrightarrow \mathbb{Z}_3$$

$$[a]_6 \longmapsto [a]_3$$

a well-defined function?

(c) Show that if $n|m$ then the rule

$$\mathbb{Z}_m \longrightarrow \mathbb{Z}_n$$

$$[a]_m \longmapsto [a]_n$$

is a well-defined function.

(d) Show that if $n \nmid m$ then the rule

$$\mathbb{Z}_m \longrightarrow \mathbb{Z}_n$$

$$[a]_m \longmapsto [a]_n$$

is *not* a well-defined function.

(9) Not well-defined.

Consider $a=1, b=4$

Note that $[a]_3 = [b]_3$ is the same element
 in $\mathbb{Z}_3$ since $a \equiv b \pmod{3}$

But $[a]_3 \longmapsto [1]_6, [b]_3 \longmapsto [4]_6$

$[1]_6$ and $[4]_6$ are different element in $\mathbb{Z}_6$.

which violates the definition of a function.

(b) well-defined

We can show that any element of $\mathbb{Z}_6$ maps to a unique element in $\mathbb{Z}_3$ by the function.

Let a, b be two arbitrary integers,
such that $[a]_6 = [b]_6$

Then $b \equiv a \pmod{6}$ by definition.

so there exists some integer k such that

$$b = a + 6k$$

so $b = a + 3(2k)$, which means

$$b \equiv a \pmod{3}$$

$$\text{then } [b]_3 = [a]_3$$

This means the same element in $\mathbb{Z}_6$ maps to only one element in $\mathbb{Z}_3$.

Therefore the function is well-defined.

(c) Let n, m be arbitrary integers with assumption that $n|m$, fix it.

Let a, b be two integers such that $[a]_m = [b]_m$.

We want to show that $[a]_n = [b]_n$, which indicates any element in $\mathbb{Z}_m$ as source maps to exactly one element in $\mathbb{Z}_n$, and this means

function is well-defined.

Since $[a]_m = [b]_m$,

$a \equiv b \pmod{m}$ by definition

so there exists some integer k such that
 $b = a + mk$.

Since $n|m$, $m = np$ for some integer p .

So $b = a + (pk)n$

Then $b \equiv a \pmod{n}$

and then $[b]_n = [a]_n$ by definition
of modular congruence.

Then we have proved the statement.

(d)

Let a be an arbitrary integer,

and consider $[a]_m$ and $[a+m]_m$

Note that $[a]_m = [a+m]_m$ since

$$a \equiv a+m \pmod{m}$$

By rule of the function, $[a]_m$ maps to $[a]_n$

while $[a+m]_m$ maps to $[a+m]_n$

Assume for sake of contradiction that

$[a+m]_n \neq [a]_n$, then there exists some
integer k such that $a+m = a + kn$, so $m = kn$,

which contradicts the condition $n \nmid m$

$$\text{so } [a+m]_n \neq [a]_n$$

Then we have one element in source mapping to multiple element in target, which violates the definition of function. $\Rightarrow$ not well-defined.

3. Let n and a be a positive integers, let $d = (a, n)$ be the gcd of a and n , and let $b \in \mathbb{Z}$. Consider the equation

$$[a]_n y = [b]_n. \quad (\star)$$

- (a) Prove that if d does not divide b , then the equation $\star$ has no solutions $y \in \mathbb{Z}_n$.
- (b) Assume $b = 0$. Show that $y = [x]_n$ is a solution to $\star$ if and only if x is multiple of $\frac{n}{d}$. Conclude that the equation $\star$ has exactly d solutions $y \in \mathbb{Z}_n$, given by the elements

$$[0]_n, \left[\frac{n}{d}\right]_n, \left[2 \cdot \frac{n}{d}\right]_n, \dots, \left[(d-1) \cdot \frac{n}{d}\right]_n.$$

Hint: It may be useful to use Theorem 1.4 of the textbook, which says that if $u, v, w \in \mathbb{Z}$, $(u, v) = 1$, and $u \mid vw$, then $u \mid w$.

- (c) Assume d divides b . Write $ra + sn = d$ for some integers $r, s \in \mathbb{Z}$. (Why is this possible?) Show that $\left[r \frac{b}{d}\right]_n$ is a solution to $\star$.
- (d) Assume d divides b . By (c) we can fix a solution $y_1 \in \mathbb{Z}_n$ to $\star$. Show that $y \in \mathbb{Z}_n$ is a solution to $\star$ if and only if $z = y - y_1 \in \mathbb{Z}_n$ is a solution to the equation

$$[a]_n z = 0.$$

Conclude that the number of solutions to $\star$ is the same as the number of solutions to $[a]_n z = 0$, and hence by (b) there are exactly d solutions to $\star$.

(a) pf. We prove it by proving its contrapositive:
if the equation has some solution $y \in \mathbb{Z}_n$,
then $d \mid b$.

Assume $y = [r]_n$ is a solution to the equation.

So $[ar]_n = [b]_n$, $ar = b + pn$ for some integer p .

So $ar - pn = b$, b is a linear combination of a, n .

Since a, n are both positive,

we can apply the conclusion on worksheet:
§ 2.2 & 2.3.;

$$\{ra + sn \mid r, s \in \mathbb{Z}\} = \{k \cdot (a, n) \mid k \in \mathbb{Z}\}$$

Since $b \in \{ra + sn \mid r, s \in \mathbb{Z}\}$,

$b \in \{k \cdot (a, n) \mid k \in \mathbb{Z}\}$, so there exists
some integer k such that $b = kd$,
that is, $d \mid b$.

Then we have proved the statement by proving
its contrapositive.

(b) Fix $b=0$.

First we prove: let $x = k \frac{n}{d}$ where k is any integer,
then $y = [x]_n$ is a solution to equation $*$.

$$\text{Since } x = k \frac{n}{d}, \quad ax = k \cdot \frac{a}{d} n$$

$$\text{Since } d = (a, n), \quad d \mid a$$

so $\frac{a}{d}$ is an integer, denote it
by q .

Then $ax = kq \cdot n$

So $[ax]_n = [0]_n, \Leftrightarrow [a]_n [x]_n = [0]_n$
 since $a, x \in \mathbb{Z}$.

Therefore $y = [x]_n$

is a solution to the equation $\star$.

Next we prove: if $y = [x]_n$ is a solution to $\star$,

then $x = k \frac{d}{n}$ for some integer k

Assume $y = [x]_n$ is a solution to $\star$,

so $ax = pn$ for some integer p

Since $d = (a, n)$, $d|a$ and $d|n$

Divide both sides of (1) by d , we have

$$\left(\frac{a}{d}\right)x = p\left(\frac{n}{d}\right) \quad (2)$$

Since $d|a, d|n$, $\frac{a}{d}$ and $\frac{n}{d}$ are all integers.

Now we claim: $\left(\frac{a}{d}, \frac{n}{d}\right) = 1 \quad (3)$

We can prove (3) by contradiction: if

$\left(\frac{a}{d}, \frac{n}{d}\right) = d \in \mathbb{Z}$ with $d \neq 1$, then
 there exists a common divisor $d \cdot d$ which
 is greater than d , violating the definition of
 greatest common divisor.

Now back to (2), denote $\left(\frac{a}{d}\right), \left(\frac{n}{d}\right)$ by u, v
 respectively and by (3) we have $(uv) = 1$

$$\text{So } ux = pv, \quad (u, v) = 1$$

By FTA, $u = \pm p_1 \dots p_g, v = \pm q_1 \dots q_h$ for some integers g, h .

And since $(u, v) = 1$, none of $p_1 \dots p_g$ is the same as any of $q_1 \dots q_h$.

So x contains all prime factors of v .

Then $x = kv$ for some integer k .

i.e. $x = k \frac{n}{d}$ for some integer k .

Then we have proved the whole statement.

This concludes that the set of solutions to equation ϕ

$$\text{is } \{ [k \frac{n}{d}]_n \mid k \in \mathbb{Z} \} = \{ [0]_n, [\frac{n}{d}]_n, [2\frac{n}{d}]_n, \dots, [(d-1)\frac{n}{d}]_n \}$$

(C) By Bezout we have $ra + sn = d$ for some integers r, s .

$$\text{So } sn = d - ra, \quad d - ra \equiv 0 \pmod{n},$$

$$[d - ra]_n = [0]_n$$

$$\Rightarrow [d]_n = [ra]_n$$

Since d divides b , $b = kd$ for some integer k .

$$\begin{aligned} \text{So } [a]_n [r \frac{d}{b}]_n &= [a]_n [rk]_n = [ark]_n \\ &= [ar]_n \cdot [k]_n = [d]_n [k]_n = [kd]_n = [b]_n \end{aligned}$$

Therefore $[r\frac{d}{b}]_n$ is a solution to equation Φ .

(d) Denote the fixed solution $y_1 \in \mathbb{Z}_n$ as $[r_1]_n$,
then $ar_1 = b + p_1n$ for some $p_1 \in \mathbb{Z}$.

Since d divides b , $b = kd$ for some $k \in \mathbb{Z}$.

First we prove: if $y \in \mathbb{Z}_n$ is a solution
to Φ , then $z = y - y_1 \in \mathbb{Z}_n$ is a solution to
the equation $[a]_n z = 0$

Assume $y = [r]_n \in \mathbb{Z}_n$ is a solution to Φ

then $[ar]_n = [b]_n$

so $ar = b + pn$ for some $p \in \mathbb{Z}$.

$$\Rightarrow a(r_1 - r) = (p_1 - p)n$$

$$\text{so } [a(r_1 - r)]_n = [0]_n$$

Therefore $[a]_n z$ is a solution to $[a]_n z = 0$

Next we prove: if $z = y - y_1 \in \mathbb{Z}_n$ is a solution to
the equation $[a]_n z = 0$, then y is a
solution to Φ .

Since $z = y - y_1$ is a solution to $[a]_n z = 0$

Denote y as $[r]_n$, $[y_1]$ as $[r_1]_n$

$$\text{Then } [a]_n [r - r_1]_n = [0]_n$$

We want to show that y is a solution to Φ ,
that is, $[a]_n [r]_n = [b]_n$

Since we know $[a]_n [r]_n = [b]_n$,

and $[a]_n [r]_n - [a]_n [r]_n = [0]_n$ by the
distribution property of modular congruence class,

$$[a]_n [r]_n = [0]_n + [b]_n = [b]_n$$

Therefore we have proved the statement.

Hence the number of solutions to Φ is the same
as the number of solutions to $[a]_n x = [b]_n$, and hence
by (b) there are d solutions to Φ .

4. Recall the notion of *equivalence relation* from the worksheet on Congruence in $\mathbb{Z}$, or look it up in Appendix B of the text.

Consider a function $f : X \rightarrow Y$ between two sets X and Y . We define a relation $\sim$ on X by saying $x \sim x'$ if $f(x) = f(x')$.

(a) Show that $\sim$ is an equivalence relation.

(b) Find a bijection between the equivalence classes on X and the image of f . Notice that this gives a partition of X .

(c) Prove that the equivalence relation on $\mathbb{Z}$ given by congruences modulo a fixed n is a particular case of the equivalence $\sim$ above: i.e., find a function f . This gives a partition of $\mathbb{Z}$; what are the equivalence classes?

(a) Pf. We show this by proving its reflexivity,
symmetry and transitivity.

Let x, y, z be three arbitrary elements in X .

$$\text{Since } f(x) = f(x) \Rightarrow x \sim x$$

$\Rightarrow$ reflexivity proved

if $x \sim y$ then $f(x) = f(y)$, so $f(y) = f(x)$,
 $\Rightarrow$ symmetry proved. $y \sim x$.

if $x \sim y$, $y \sim z$ then $f(x) = f(y)$, $f(y) = f(z)$
 $\Rightarrow f(x) = f(z)$
 $\Rightarrow$ transitivity proved. $\Rightarrow x \sim z$

(b) We can define the equivalence class $[y]$ as

$$\{x \in X \mid f(x) = y\}$$

And we define the set of all such equivalence classes on X as X_f .

$$\text{So } X_f = \{[y] \mid y \in \text{Im}(f)\}$$

And finally we define a function

$$\varphi: X_f \rightarrow \text{Im}(f)$$

as $[y] \mapsto y$.

We can show that:

(1) it is well-defined.

Let $[y]$ be an arbitrary equivalence class.
under our definition

$\forall x \in [y]$, $f(x) = y$, so it maps to exactly

one element in Y .

② it is injective:

Let y_1, y_2 be two arbitrary elements in $\text{Im}(f)$
we want to show $y_1 = y_2 \Rightarrow [x_1] = [x_2]$

Assume $x_1 = x_2$, and then assume for contradiction that $[x_1] \neq [x_2]$

so there exists $x_1 \in [x_1]$ s.t. $f(x_1) \neq y_2$,

Then we have proved injectivity. then $f(x_1) \neq y_1$,
contradicts.

③ it is surjective.

Since $\forall y \in \text{Im}(f), \exists x \in X$ s.t. $f(x) = y$

by definition of image
Consider the congruence class of this x :

So any element in $\text{Im}(f)$ is $[y]$.

mapped by at least one element

in X_f , therefore it is surjective.

④. By ①, ②, ③, it is bijective.

(C) Pf Consider the function $f: \mathbb{Z} \rightarrow \mathbb{Z}_n$
$$x \mapsto [x]_n.$$

So the relation $x \equiv x' \pmod{n}$

is just $(x \sim x' \text{ if } f(x) = f(x'))$

by definition of congruence modulo class.

This function is well-defined since if we take $x=y$, then $f(x) = [x]_n = [y]_n = f(y)$

Therefore the equivalence relation on $\mathbb{Z}$ given by congruence modulo a fixed n is a particular space.

And by (b), we can conclude that f gives a partition of $\mathbb{Z}$.

The equivalence classes are $[0]_n, [1]_n, \dots, [n-1]_n$.